SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	HARIHARAN G	**Reg. No.:**	192424156
Section / Batch:	B.Tech AIDS / 3rd Year	**Date:**	29/08/2026

Code of Conduct

I HARIHARAN G(192424156) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________HARIHARAN G_________

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression Possible Antecedents Valid/Invalid Reason

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. **Maintain entity coherence using assignment, mistakes, and you.**
2. **Use a Cause–Effect or Elaboration discourse relation.**
3. **Include at least two keywords from:**
    - **review**
    - **improve**
    - **confident**
    - **feedback**
4. **Response length must be 2–3 sentences.**

5. **Maintain a positive and polite tone.**
6. **Do not give contradictory advice.**

Tasks:

a) Identify the dialog acts required.

b) Identify the important entities that must be maintained.

c) Generate three possible responses satisfying all constraints.

d) Compare the responses based on:

- **Coherence**
- **Constraint satisfaction**
- **Politeness**
- **Relevance**

e) Select the best response and justify your answer.

f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- **A financial institution**
- **The side of a river**

Constraints:

1. **Resolve the correct word sense using contextual information.**
2. **Identify important entities and actions.**
3. **Represent the meaning using predicate logic.**
4. **Preserve semantic relationships.**
5. **Maintain coherence in the generated paraphrase.**

Tasks:

a) Identify the possible meanings of bank.

b) Select the correct sense and justify your decision using contextual constraints.

c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	4	4	4	4	3	19	97
2	4	4	4	4	4	20	
3	4	4	4	4	4	20	

Faculty In-charge	Course Coordinator	Program Director
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Signature: ________________	Signature: ________________	Signature: ________________
Date: 29/08/2026	Date: ____________________	Date: ____________________

1. Reference Resolution and Discourse Coherence

a) Referring Expressions and Possible Antecedents

The important referring expressions are:

- **He**
- **him**
- **it**
- **they**
- **the book**

Possible antecedents are **Ravi**, **Arun**, and **a book**.

b & c) Reference Resolution Table

Referring Expression	**Possible Antecedents**	**Valid/Invalid**	**Reason**
He	Ravi, Arun	Ravi – Valid; Arun – Valid	Both are singular male entities; recency favors Arun, but discourse context is needed
him	Ravi, Arun	Ravi – Valid; Arun – Invalid	Arun is already the helper/subject of the sentence, so “him” naturally refers to the person being helped: Ravi
it	book, Ravi, Arun	Book – Valid; others – Invalid	“find it” semantically refers to an object, namely the book
they	Ravi, Arun	Valid	Plural pronoun refers to both people
the book	book	Valid	Refers back to the previously introduced book

Applying the Constraints

Gender and number agreement:

“he” and “him” must refer to a singular male person. “They” refers to the two people.

Recency:

The most recently mentioned compatible entity is generally preferred. However, recency alone is not sufficient.

Grammatical role:

In “Arun helped him,” Arun is the subject and “him” is the object. This supports **him = Ravi**.

Semantic compatibility:

A person can look for and discuss a book, while a book cannot help a person. Therefore, “him” must refer to a person and “it” to the book.

Discourse coherence:

The paragraph describes Ravi and Arun performing related actions around the same book, making the following chains coherent.

d) Final Coreference Chains

Person chain 1:

Ravi → He → him

Person chain 2:

Arun

Person group:

Ravi + Arun → they

Object chain:

a book → it → the book

e) Rewritten Paragraph

Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.

f) Ambiguity if "He was looking for a book" Appears First

If the sentence **"He was looking for a book"** appeared before Arun was introduced, there would be no suitable antecedent for **He**. The reader would not know who "He" refers to.

For example:

"Ravi arrived at the library. He was looking for a book. Arun met Ravi later."

Here, "He" can reasonably refer to Ravi because Ravi has already been introduced.

If neither Ravi nor Arun had been introduced, the pronoun would have an **unresolved reference**, reducing discourse coherence.

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2. Dialog Act Recognition and Response Generation

Conversation

Student:

"I submitted my assignment, but I am worried that I may have made many mistakes."

The required dialog acts are **Reassure + Advise**.

a) Required Dialog Acts

1. **Reassure** – Reduce the student's concern and provide encouragement.
2. **Advise –** Suggest a useful action, such as reviewing the assignment and using feedback to improve it.

b) Important Entities

The response should maintain these entities:

- **assignment**
- **mistakes**
- **you** – the student

Maintaining these entities ensures that the response remains connected to the student's original statement.

c) Three Possible Responses

It's normal to worry about mistakes after submitting an assignment, so don't be too hard on yourself. You can review the feedback carefully and use it to improve your future work, which will help you feel more confident.

d) Comparison

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
Supportive	High	High	High	High
Encouraging	High	High	High	High
Practical	Very High	High	High	Very High

All three responses maintain the assignment/mistakes context and contain the required keywords. They also use a cause–effect relationship, such as **reviewing feedback → improving work → becoming more confident**.

e) Best Response

The **Practical** response is the best because it directly addresses the student's worry while providing a clear and realistic suggestion. It maintains entity coherence, uses the required keywords, remains positive and polite, and provides a clear cause–effect relationship.

f) Effect of Losing Entity Coherence

If entity coherence is not maintained, the response may suddenly discuss unrelated topics or refer to unclear objects.

For example:

"Don't worry about your assignment. Review the doctor's feedback and improve your health."

The response shifts from **assignment** to **doctor/health**, which does not match the conversation. This makes the chatbot response confusing and irrelevant.

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3. Word Sense Disambiguation and Meaning Representation

Source Sentence

"Priya sat on the bank and watched the boats moving across the water."

The word **bank** has multiple meanings.

a) Possible Meanings of "Bank"

1. **Financial institution** – a place where people deposit or withdraw money.
2. **River bank** – the land beside a river or body of water.

b) Correct Sense

The correct sense is:

bank = side of a river

Justification

The surrounding words provide strong contextual evidence:

- **sat on the bank**
- **boats**
- **water**
- **moving across the water**

A person can sit beside a river and watch boats on the water. A financial institution does not fit naturally with boats and water.

Therefore:

$$\boxed{\text{bank = river bank}}$$

This is an example of **Word Sense Disambiguation (WSD)**, where contextual information is used to select the correct meaning of an ambiguous word.

c) Predicate Logic Representation

Let:

- Priya = p
- Bank = b
- Boat = z
- Water = w

A suitable representation is:

$$sit(p)$$
$$bank(b)$$
$$beside(p, b)$$
$$watch(p, z)$$
$$boat(z)$$
$$move(z)$$
$$water(w)$$

The complete representation can be written as:

$$\exists p, b, z, w[sit(p) \land bank(b) \land beside(p, b) \land watch(p, z) \land boat(z) \land move(z) \land water(w)]$$

A more specific representation can include the relationship between the boats and water:

$$\exists p, b, z, w[Priya(p) \land RiverBank(b) \land sit(p) \land beside(p, b) \land boat(z) \land watch(p, z) \land move(z) \land water(w) \land moveAcross(z, w)]$$

This preserves the semantic relationships between **Priya, bank, boats, and water**.

d) Unambiguous English Paraphrase

Priya sat beside the river and watched the boats moving across the water.

The word **bank** has been replaced with **beside the river**, making the intended meaning clear.

e) How WSD Improves Machine Translation

Word Sense Disambiguation helps machine translation systems select the correct meaning of an ambiguous word before translating it.

For example:

"She sat on the bank and watched the boats."

Without WSD, a translation system might incorrectly interpret **bank** as a financial institution.

With WSD, the surrounding words **sat, boats, and water** indicate that **bank** means the **side of a river**.

Therefore, WSD:

- Reduces translation errors.
- Preserves the original meaning.
- Improves contextual accuracy.
- Helps select appropriate translated words.
- Produces more natural and coherent translations.

Overall Conclusion

Reference resolution maintains connections between entities across sentences, dialog act recognition helps conversational systems generate appropriate responses, and WSD resolves ambiguous word meanings using context. Together, these NLP techniques improve **coherence, semantic understanding, and accuracy** in intelligent language-processing systems.